

$$w_1^{(6)}(u_{12} \otimes f_h) = 0 \text{ for } h = 1, 2, 6, \quad w_1^{(6)}(u_{12} \otimes f_h) = b_{12h} \text{ for } h = 3, 4, 5,$$

$$w_1^{(6)}(u_{1234} \otimes f_h) = 0 \text{ for } h \neq 5, \quad w_1^{(6)}(u_{1234} \otimes f_5) = b_{12345}.$$

To compute the next map we use the fact that for $1 \leq h \leq 5$, $w_1^{(6)}(u_H \otimes f_h)$ is nonzero if and only if $h \notin H$. This gives

$$q_2^{(6)}(f_1) = b_1 b_{12345} f_6 + b_{12345}(f_1 + b_1 f_6) + P_1 f_6 + w_2^{(1)}(\hat{e}_1) = (b_1 b_{12345} + 2P_1) f_6,$$

and similarly for other indices $\neq 6$. Thus we obtain

$$w_2^{(6)}(f_h) = b_1 b_{12345} + 2P_h \quad \text{for } 1 \leq h \leq 5 \quad \text{and} \quad w_2^{(6)}(f_6) = -b_{12345}.$$

The ideal generated by the entries of $w_2^{(6)}$ is $(P_1, \dots, P_5, b_{12345})$. It is a hyperplane section of the ideal of submaximal pfaffians of the skew-symmetric matrix defined by the variables b_{ijk} . The matrix of $w_2^{(1)}$ is a presentation matrix of this ideal. On the other hand, the top component of the critical representation W_0 contains information about the spinor structure of this related ideal.

The computation of $w_{1,1}^{(0)}$ is straightforward. Using the formulas for $w_0^{(0)}$ we simply get $w_{1,1}^{(0)}(1) = b_{12345}$. Also for $w_{1,2}^{(0)}$ we easily obtain $w_{1,2}^{(0)}(\theta_{ij}) = w_{1,2}^{(0)}(\theta_{\hat{i}j}) = w_{1,2}^{(0)}(\theta_{i\hat{j}}) = 0$ and $w_{1,2}^{(0)}(\theta_{\hat{i}\hat{j}}) = b_H$ where $H = \{1, 2, 3, 4, 5\} \setminus \{i, j\}$. For $w_2^{(0)}$, if $|H| = 3, 5$ we get $q_2^{(0)}(u_h) = b_{12345} \cdot w_1^{(1)}(u_I)$ and $w_2^{(0)}(u_H) = b_H b_{12345}$. For $H = \{i\} \subset \{1, \dots, 5\}$, we get

$$q_2^{(0)}(u_i) = w_2^{(1)}(\hat{e}_i) + (f_i + b_i f_6) b_{12345} = P_i f_6$$

and consequently $w_2^{(0)}(u_i) = P_i$. We will shortly see that the entries of $w_2^{(0)}$ give a choice of spinor coordinates, i.e. a choice of the structure map $w_0^{(0)}$, for the minimal free resolution of the ideal $(P_1, \dots, P_5, b_{12345})$.

2.4 The minimal free resolution of a hyperplane section of pfaffians

Let us now analyze higher structure maps in the case of the complex $\mathbb{F}$ which is the minimal free resolution of a hyperplane section of the ideal generated by submaximal pfaffians of a 5×5 skew-symmetric matrix. Define the skew-symmetric matrix on variables $x_{ij} = -x_{ji}$ and denote its pfaffians using the notation

$$P_{\hat{1}} = x_{23}x_{45} - x_{24}x_{35} + x_{25}x_{34},$$

with the other pfaffians obtained from this one by permutation of indices. Let y be an element of R regular modulo the ideal $(P_{\hat{1}}, \dots, P_{\hat{5}})$. The complex $\mathbb{F}$ resolving the Gorenstein ideal $(P_{\hat{1}}, \dots, P_{\hat{5}}, y)$ has format $(1, 6, 10, 6, 1)$. We consider bases on F and G such that the